FINAL SCAN SHEET

Every ★ HIGH YIELD (14) and ⚠ TRAP (8) from the guide — nothing else. Five minutes, closed book, right before you walk in.

★ HIGH YIELD — things he explicitly flagged

1. Lymph is also connective tissue. He said "blood and also lymph" — a matching question could pair either one.

2. The tube *is* the argument. RBCs occupy 45% of blood volume; every WBC and platelet in your body fits in the <1% buffy coat. That's why RBC count — and nothing else — sets viscosity.

3. Plasma proteins are NOT used by cells as fuel. He lingered here. RBCs and WBCs sit directly in a protein-rich fluid and never touch it — they use glucose, fatty acids, O_2. This is a ready-made true/false question.

4. The RBC carries oxygen but never uses it. He called this the elegant part of the design — the cell responsible for all the oxygen doesn't touch it. Lacking mitochondria is *why*. Expect this as a "which is true" item.

5. "Once committed to a pathway, they cannot change." A myeloid progenitor can never become a lymphocyte. He said this almost verbatim — it's phrased like a true/false item.

6. EPO comes from the KIDNEYS (liver to a small extent), and **hypoxia** is the trigger. Kidney → EPO → red marrow → more RBCs → O_2 restored → stimulus off.

7. WBCs are the only TRUE cells among the formed elements — nucleus plus every organelle. RBCs have no nucleus; platelets aren't even whole cells. He stressed this phrasing, and "true cell" is precise exam language.

8. Eosinophilia — he named this as licensing-exam material. Eosinophils above the normal 2–4% points to **parasitic infection, allergies, or asthma**. Think eczema, allergic rhinitis, seasonal allergy — and note it may only be raised *during* a reaction, not permanently.

9. Platelets are not cells. They are **cytoplasmic fragments** broken off a much larger marrow cell, the **megakaryocyte** (mega = large). No nucleus → they age fast → removed within ~10 days if unused.

10. Vascular spasm is "probably the most important factor" — his words — because it is immediate and cuts flow to the injury before anything else can happen. But it is never sufficient alone; a small leak persists, which is why steps 2 and 3 exist.

11. Press and hold. Never massage. The platelet plug and fibrin mesh form within the first minute and are **still fragile**. Massage tears them → re-bleeding → and here's the detail he added: **you may see nothing at the time and a big bruise the next day.** Press ~30 seconds; once bleeding stops, you're done.

12. Thrombus vs embolus is a definition pair, and the difference is motion. Thrombus = forms and **persists in an unbroken vessel** — no trauma. Embolus = the same clot once it **travels**. DVT is the classic thrombus.

13. Why does liver failure cause bleeding? He asked this deliberately, and a student guessed erythropoietin (wrong — that's kidney). The chain: **clotting factors are proteins → the liver makes proteins → severe liver disease → no fibrinogen or prothrombin → bleeding.** This connects straight back to §2. Expect it.

14. Newborns get a vitamin K injection at birth. Vitamin K is made by **gut bacteria**, and a newborn's gut hasn't been colonised yet. It's prophylaxis — so that any injury doesn't become fatal.

⚠ TRAP — misconceptions he corrected in class

1. Thick blood does not cause atherosclerosis. A student asked this directly; he said atherosclerosis is **plaque buildup** (lipid). Viscosity affects how easily blood *moves*; plaque is a separate mechanism. Don't let a question merge them.

2. A high reticulocyte count is not a disease. A student asked; he was firm: it is a **compensatory mechanism**. The marrow is responding correctly to a problem elsewhere — bleeding, hemolysis, or iron finally arriving. The pathology is somewhere else.

3. Thyroid does not directly drive erythropoiesis. Another student asked. Thyroid sets *cellular metabolic rate*; any link to RBC production is **indirect**.

4. Stress alone can raise your WBC. He said that when a count sits at the high end, he asks about the day of the draw — a stressful day can naturally double it. A high WBC is not automatically infection.

5. WBC count alone CAN indicate sepsis. A student asked whether you must wait for culture. His answer: no — “definitely alone can be indicator.” Sepsis kills in hours via blood pressure collapse, so clinicians start **broad-spectrum antibiotics immediately** and let culture refine it later.

6. NK cells DO have granules — they're classed as agranulocytes only because the granules aren't visible under light microscopy. He raised this unprompted. Also note he corrected a student who thought all five types were granulocytes: **lymphocytes and monocytes are agranulocytes.**

7. A student asked whether more nitric oxide is therefore better. His answer: **no — the normal amount**. “Nothing to say the higher is the better or the lower is the better. Just always the normal amount.” Beware any option phrased as “more X is better.”

8. Aspirin does not destroy platelets, and no blood thinner touches RBCs. He was explicit: “not destroying platelets, just the ability for aggregation kind of reduces.” The count stays normal; the stickiness drops.

If any of these don't ring a bell instantly — go find it in the full guide tonight, not tomorrow at 6:30.